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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte(Deemed to be University)
First Semester M.Tech. (CCT)(CBCS) Degree Examinations
January 2023

22CCT111- SPECIAL CONCRETES

Duration: 3 Hours

Max. Marks:100

- Note:** 1) Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.
2) Use of IS456-2000 and IS-10262-2019 is permitted.
3) Assume missing data suitably.

		Marks	BT*	CO*	PO*
Unit – I					
1.	Design mix proportions of concrete. The following data are given: Specified 28-day compressive strength: 30 MPa Assume missing data suitably.	20	L*6	1	1,3
2.	Design mix proportions of heavy weight concrete to arrive M-10 mix.	20	L6	2	3,5
3.	a) Explain the properties and applications of Rice Husk ash and Silica Fume in concrete.	10	L2	1	2,4
	b) Explain Characteristics of LWC. Specify the applications and limitations of the same in Civil Engineering.	10	L3	2	1,2
Unit – II					
4.	Design SCC for target strength of 28 MPa, Assuming all necessary data suitably.	20	L6	3	3
5.	Design M90 Grade bendable concrete. Assume necessary data suitably.	20	L6	4	1
6.	a) Describe the various lab tests conducted for SCC along with its criteria.	10	L2	3	3
	b) Describe the special properties of ECC, compared to normal FRC and normal HFRCC	10	L2	4	1
Unit – III					
7.	a) Design the mix proportion for M-105 Grade concrete assuming all necessary data.	20	L6	5	3
	b) What is RMC Explain various types of RMC along with advantages and disadvantages.	20	L2	5	3

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
